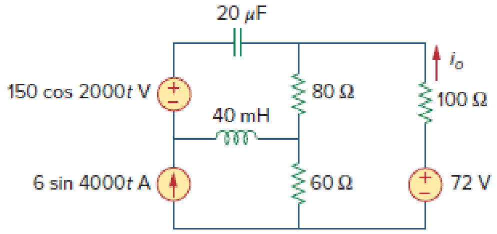
Chapter 10, Problem 48P

Problem

Find *io* in the circuit of Fig. 10.93 using superposition.

Figure 10.93:



Step-by-step solution

Step 1 of 14

Refer to Figure 10.93 in the textbook.

Case 1:

Consider the current source of $6\sin 4000t$ as active and remove remaining sources by short circuiting the voltage sources.

Here, $\omega = 4000 \text{ rad/s}$

Draw the modified circuit diagram.

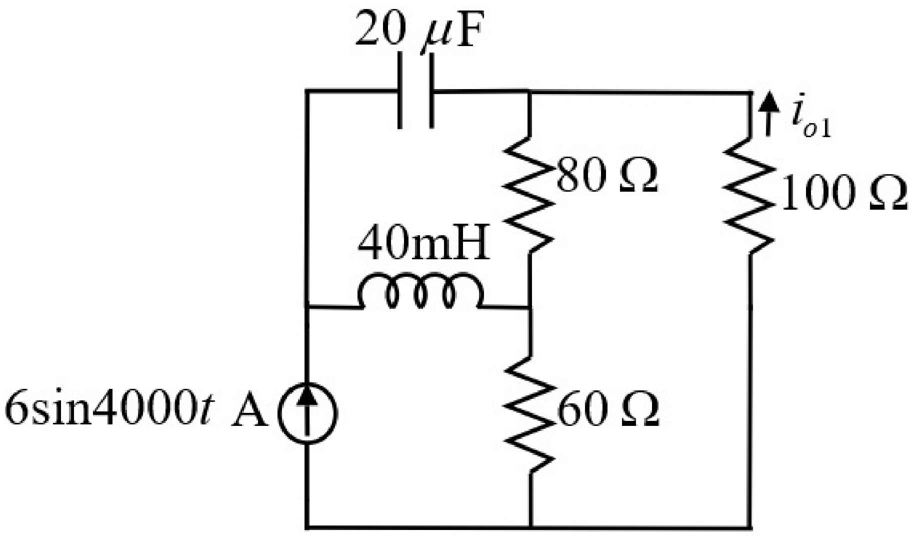


Figure 1

Step 2 of 14

Calculate the impedance of the capacitor.

$$\begin{aligned}\mathbf{Z}_C &= \frac{1}{j\omega C}\\ &= \frac{1}{j(4000)(20\times10^{-6})}\\ &= -j12.5\ \Omega\end{aligned}$$

Calculate the impedance of the inductor.

$$\begin{aligned}\mathbf{Z}_L &= j\omega L\\ &= j(4000)(40\times10^{-3})\\ &= j160\ \Omega\end{aligned}$$

Step 3 of 14

Draw the circuit diagram in frequency domain.

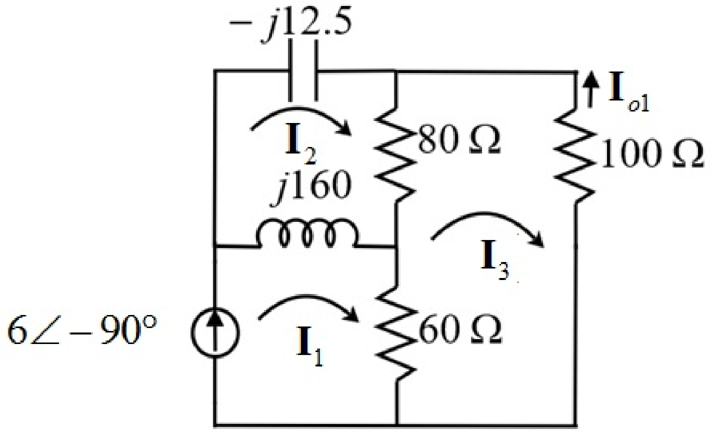


Figure 2

Step 4 of 14

From Figure 2, the value of current $\mathbf{I}_1$ is,

$$\begin{aligned}\mathbf{I}_1 &= 6\angle-90°\\ &= -j6\ \text{A}\end{aligned}$$

Apply the Kirchhoff's voltage law in loop 2.

$$-j12.5\mathbf{I}_2 + 80(\mathbf{I}_2 - \mathbf{I}_3) + j160(\mathbf{I}_2 - \mathbf{I}_1) = 0$$
$$-j160\mathbf{I}_1 + (80 + j147.5)\mathbf{I}_2 - 80\mathbf{I}_3 = 0$$

Substitute $-j6$ for $\mathbf{I}_1$.

$$(-j160)(-j6)\mathbf{I}_1 + (80 + j147.5)\mathbf{I}_2 - 80\mathbf{I}_3 = 0$$

$$(80 + j147.5)\mathbf{I}_2 - 80\mathbf{I}_3 = 960 \quad \text{......} \ (1)$$

Step 5 of 14

Apply the Kirchhoff's voltage law in the loop 3.

$$60(\mathbf{I}_3-\mathbf{I}_1)+80(\mathbf{I}_3-\mathbf{I}_2)+100\mathbf{I}_3=0$$
$$-60\mathbf{I}_1-80\mathbf{I}_2+240\mathbf{I}_3=0$$

Substitute $-j6\text{ A}$ for $\mathbf{I}_1$.

$-80\mathbf{I}_2+240\mathbf{I}_3=-j360$ ...... (2)

Solve the equations (1) and (2).

$$\mathbf{I}_2=1.3617-j6.016\text{ A}$$

$$\mathbf{I}_3=0.4539-j3.505\text{ A}$$

Step 6 of 14

Calculate the value of the current, $\mathbf{I}_{o1}$.

$$\begin{aligned}\mathbf{I}_{o1}&=-\mathbf{I}_3\\&=-(0.4539-j3.3505)\\&=-0.4539+j3.3505\\&=3.38\angle 97.71^\circ\text{ A}\end{aligned}$$

Calculate the value of the current in time domain.

$$i_{o1}=3.38\sin(4000t+97.71^\circ)\text{ A}$$

Comments (1)

Anonymous

Should be 3.38sin(4000t+187.38)A Convert angle 97.71° from cos to sin (+90º)

Step 7 of 14

Case 2:

Consider the voltage source of $150\cos 2000t$ as active and remove remaining sources by short circuiting the voltage sources and open circuiting the current sources.

Here, $\omega=2000\text{ rad/s}$

Calculate the impedance of the capacitor.

$$\begin{aligned}\mathbf{Z}_C&=\frac{1}{j\omega C}\\&=\frac{1}{j(2000)(20\times10^{-6})}\\&=-j25\ \Omega\end{aligned}$$

Calculate the impedance of the inductor.

$$\begin{aligned}\mathbf{Z}_L&=j\omega L\\&=j(2000)(40\times10^{-3})\\&=j80\ \Omega\end{aligned}$$

Step 8 of 14

Draw the circuit diagram in frequency domain.

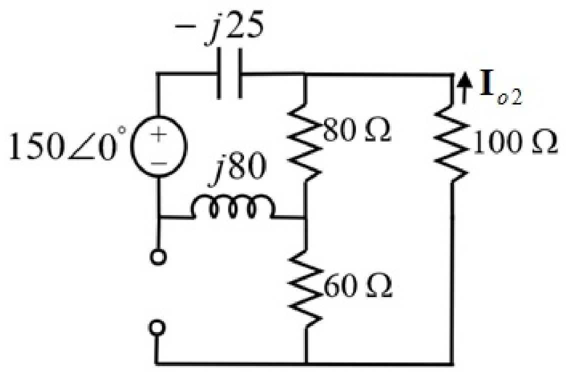


Figure 3

Step 9 of 14

In Figure 3, resistance $100\ \Omega$ and $60\ \Omega$ are in series. The equivalent resistance is,

$$R_1 = 100 + 60$$
$$= 160\ \Omega$$

The equivalent resistance $R_1 = 160\ \Omega$ and $80\ \Omega$ are in parallel. The equivalent resistance is,

$$R_2 = \frac{160 \times 80}{240}$$
$$= 53.33\ \Omega$$

Step 10 of 14

Draw the modified circuit diagram.

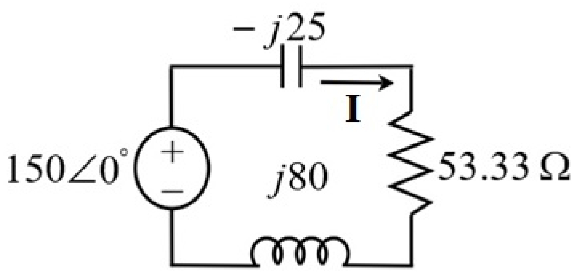


Figure 4

Step 11 of 14

Calculate the value of current, $\mathbf{I}$.

$$\begin{aligned}\mathbf{I} &= \frac{150\angle 0^\circ}{53.33 + j55} \\ &= -1.363 - j1.4 \\ &= 1.958\angle -45.88^\circ \text{ A}\end{aligned}$$

Apply current division rule to calculate the current, $\mathbf{I}_{o2}$ in Figure 3.

$$\begin{aligned}\mathbf{I}_{o2} &= -\left(1.958\angle -45.88^\circ\right)\left(\frac{80}{80+160}\right) \\ &= \left(1.958\angle -45.88^\circ\right)\left(\frac{-1}{3}\right) \\ &= 0.6526\angle 134.1^\circ \text{ A}\end{aligned}$$

Calculate the value of the current in time domain.

$$i_{o2} = 0.6526\cos\left(2000t + 134.1^\circ\right) \text{ A}$$

Step 12 of 14

Case 3:

Consider the voltage source of 72 V and remove remaining sources by short circuiting the voltage sources and open circuiting the current sources.

For DC sources, the capacitor acts as open circuit and inductor acts as short circuit.

Draw the modified circuit diagram.

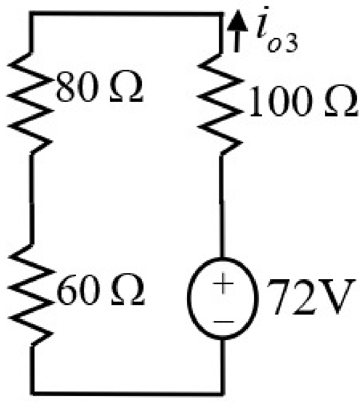


Figure 5

Step 13 of 14

Calculate the value of the current, i_{o3}.

$$\begin{aligned}i_{o3} &= \frac{72}{60+80+100} \\ &= \frac{72}{240} \\ &= 0.3 \text{ A}\end{aligned}$$

Step 14 of 14

Apply superposition theorem to calculate the output current.

$$
\begin{aligned}
i_o &= i_{o1} + i_{o2} + i_{o3} \\
&= 3.38\sin\left(4000t + 97.71°\right) + 0.6526\cos\left(2000t + 134.1°\right) + 0.3 \\
&= 0.3 + 3.38\sin\left(4000t + 97.71°\right) + 0.6526\cos\left(2000t + 134.1°\right)\text{ A}
\end{aligned}
$$

Thus, the expression for the current i_o is,

$\boxed{0.3 + 3.38\sin\left(4000t + 97.71°\right) + 0.6526\cos\left(2000t + 134.1°\right)\text{ A}}$.